

Bayesian statistics is an approach to data analysis and parameter estimation based on Bayes Theorem.

Bayes Theorem

Probability $\begin{cases} \rightarrow \text{Independent Event} \\ \rightarrow \text{dependent Event} \end{cases}$

i) Independent Event

Rolling a die

$\{1, 2, 3, 4, 5, 6\}$

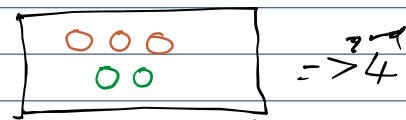
$$P(1) = \frac{1}{6} \quad P(2) = \frac{1}{6}$$

Tossing a coin

H, T

$$P(H) = 0.5 \quad P(T) = 0.5$$

ii) Dependent Event



if I pick a ball

$$P(X = \text{green}) = \frac{2}{5} \Rightarrow 1^{\text{st}}$$

$$P(X = \text{orange}) = \frac{3}{4} \Rightarrow$$

$$P(X = \text{green}) = \frac{1}{3} \Rightarrow$$

$$P(X = \text{green}) = P(X) \cdot P(Y|X)$$

$$P(A \text{ and } B) = P(A) \cdot P(B|A)$$

$\Downarrow$

Conditional probability

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$$

$$P(B|A) = \frac{P(B) \cdot P(A|B)}{P(A)}$$

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)} \Rightarrow \text{Bayes Theorem}$$

$$P(A/B) = \frac{P(A) \cdot P(B|A)}{P(B)} \Rightarrow \text{Bayes Theorem}$$

Terms when

$A, B = \text{Event}$

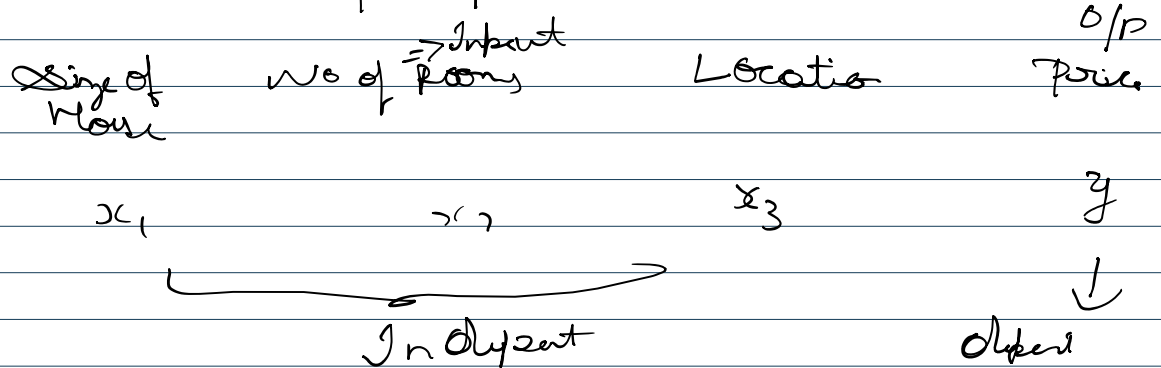
$P(A/B) = \text{Probability of A given B is true}$

$P(B/A) = \text{Probability of B given A is true}$

$P(A), P(B) = \text{Independent probability of A \& B}$

Ex :-

Dataset $\rightarrow$ House price prediction



$$P(y/x_1 x_2 x_3) = \frac{P(y) * P(x_1 x_2 x_3 | y)}{P(x_1 x_2 x_3)}$$